



S.A. ENGINEERING COLLEGE

(Autonomous - Institute Level Research Centre, Affiliated to Anna University, Chennai)

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Domain name : Software

Problem Statement Title: Rural Awareness and Accessibility of Tamil Nadu
Government Schemes (Problem 35)

Team Name: Static Squad

Team Head: Jashwanth G

Project Title: Thitta-Murasu (திட்ட முரசு - The Scheme Announcer)





TITLE OF THE IDEA & ABSTRACT

- The Idea: "Thitta-Murasu" is a centralized, offline-capable dashboard designed exclusively for Panchayat officials to proactively identify and notify eligible scheme beneficiaries.
- How it addresses the problem: It bypasses the rural digital divide. Instead of expecting unconnected citizens to search for schemes, the local government uses this tool to push information directly to them.
- Innovation & Uniqueness: Reverses the standard model from "citizen-seeking-scheme" to "scheme-seeking-citizen." It requires zero internet, smartphone, or digital literacy from the rural public.





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TECHNICAL APPROACH

Technologies: Frontend: React.js (Offline-first PWA dashboard)

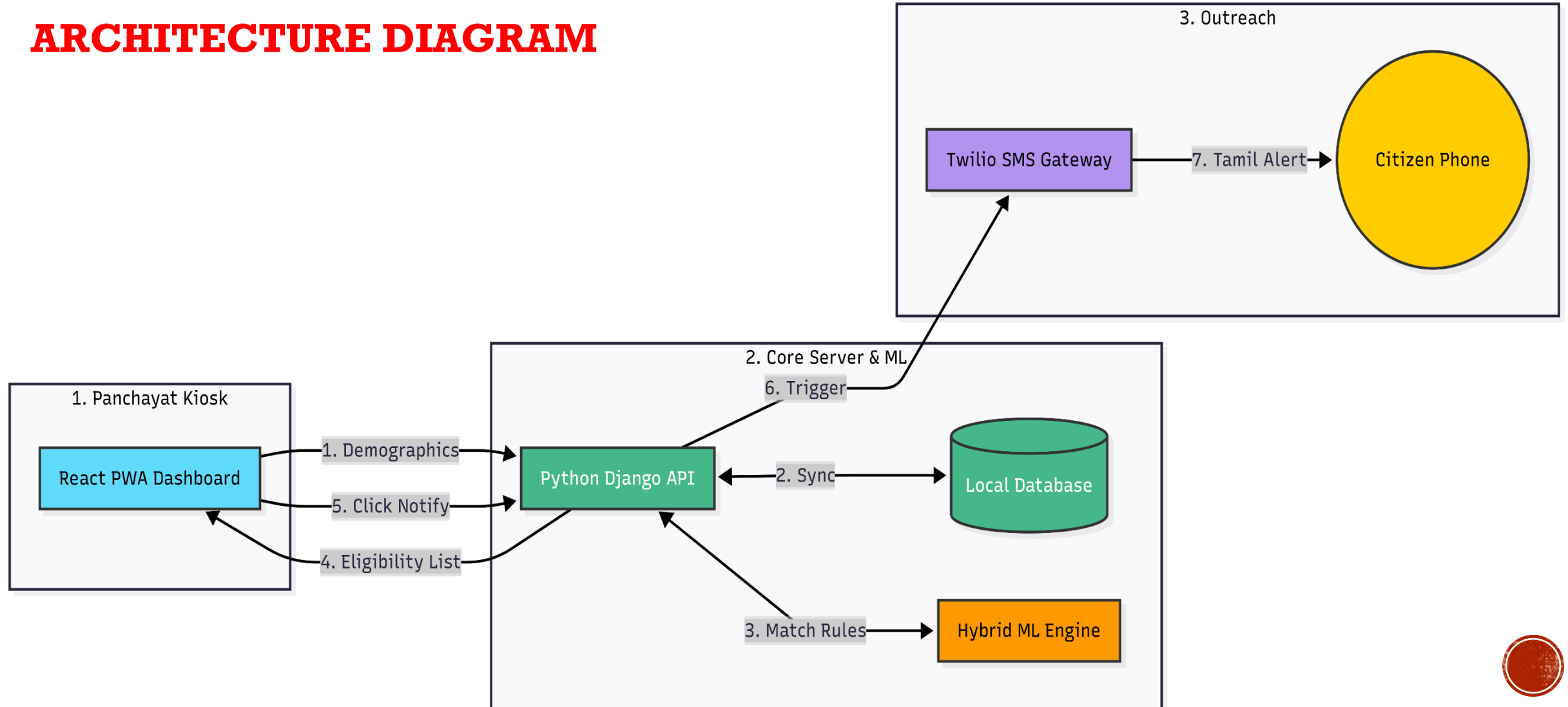
- Backend: Python/Django
- AI/ML Engine: Scikit-Learn (Hybrid Model)
- Comms: SMS Gateway API (Twilio/Local provider)

Methodology:

- Official inputs citizen demographics (Age, Income, Caste, SHG status).
- Hybrid ML Engine filters data against 100+ TN scheme criteria with dashboard outputs exact matched eligible schemes.
- 1-Click trigger sends an automated Tamil SMS to the citizen's feature phone with application instructions.



ARCHITECTURE DIAGRAM





FEASIBILITY AND VIABILITY

- **Feasibility Analysis:** Highly viable. Software runs smoothly on existing Panchayat/e-Sevai computers. Processes lightweight tabular data requiring very low computing power.
- **Potential Challenges:** Inaccurate or outdated citizen mobile numbers in the database.
- Data privacy when handling demographic information.
- **Strategies to Overcome:** Implement strict Role-Based Access Control (RBAC) for official logins.
- Enable a "Print" feature to generate physical Eligibility Slips that officials can hand-deliver when SMS fails.





IMPACT AND BENEFITS

- **Target Audience Impact:** Achieves 100% digital inclusion. Reaches the most vulnerable populations (BPL, elders) without them needing to navigate complex government websites.
- **Social Benefits:** Maximizes the utilization of critical welfare programs (housing, education, maternal health) at the grassroots level.
- **Economic Benefits:** Prevents state fund underutilization and injects financial aid directly into the rural village economy.
- **Administrative Benefits:** Drastically reduces manual rulebook-checking and workload for local officers.





RESEARCH & REFERENCES

- **TN Government Frameworks:** Guidelines from the Tamil Nadu State Rural Livelihoods Mission (TNSRLM) and Participatory Identification of Poor (PIP) manual.
- **Connectivity Data:** Telecom Regulatory Authority of India (TRAI) rural reports highlighting high SMS delivery rates vs. low 4G consistency.
- **Technical Research:** Methodologies on deploying Hybrid Machine Learning Systems for high-accuracy demographic filtering.
- **Offline-First Software Architecture:** Research on Progressive Web App (PWA) local caching mechanisms and decentralized databases to ensure uninterrupted software operation during network outages at local Panchayat offices.
- **Vernacular e-Governance Communication:** Technical frameworks for integrating regional SMS gateways (e.g., C-DAC Mobile Seva or localized Twilio APIs) to ensure accurate, automated Tamil text delivery to non-smartphones.





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Thank you

